Charlie Atkinson

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Personal Profile

Robotics and AI graduate from the University of Hull with a passion for Artificial Intelligence, Machine Learning, and Data Science. Skilled in Python, machine learning algorithms, computer vision, and NLP, with practical experience through projects and self-directed study. Seeking a graduate role in Data Science to further develop technical expertise.

Technical Skills

Programming: Skilled in Python and C# with experience in JavaScript, CSS, HTML, SQL and Arduino.

Machine Learning & Data Science: TensorFlow, Scikit-learn, NumPy, Pandas, NLTK, Matplotlib

Software & Tools: Jupyter Notebook, Arduino IDE, MATLAB, Simulink, Blender, Excel, Ubuntu, ROS2

Key Competencies: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Visualisation.

Education

University of Hull, UK **Sept 2023 – July 2026**

BSc Robotics and Artificial Intelligence

- Engineered a real-time computer vision referee using YOLOv11 Nano and OpenCV, achieving 0.970 mAP@50 and 12–15 FPS on CPU while implementing a temporal debounce filter and deterministic state machine to autonomously validate card plays and flag cheating.
- Developed an autonomous multi-agent pursuit simulation in ROS 2 Humble, implementing decentralised inter-node communication, predictive lead-pursuit ($0.4\,\text{s}$ lookahead), and virtual repulsion fields across 4 agents to achieve a First-Class grade (1st).
- Achieved top grades across core modules, including Artificial Intelligence and Data Analysis & Visualisation, demonstrating strong technical proficiency in machine learning and algorithmic problem-solving.
- Built complex AI systems and search models, implementing A* search, DFS, BFS, Dijkstra, and Q-learning for smart city logistics and reinforcement learning tasks.
- Conducted end-to-end predictive modelling and fairness evaluation, applying Decision Trees, XGBoost, ensemble methods, and K-medoids clustering with interpretability via SHAP and LIME.
- Developed embedded robotics and .NET applications, including sensor-driven autonomous robot cars, RFID access-control systems, and C# console applications.
- Designed and pitched a robotics solution for the Professional Development module, collaborating in an agile team to solve a real-world automation problem and achieving a First-Class grade (92%).

Relevant Work Experience

Virgin Media O2 (UK) **Summer 2024**

- Tasked with completing a data science interview project involving cleaning, analysing and processing a large dataset.
- Evaluated different machine learning models and decided on a Random Forest Classifier.
- Trained and tuned the model, evaluated performance using confusion matrices and F1 score, improving accuracy.
- Enhanced my skills in Jupyter Notebook, Scikit-learn, Matplotlib and Microsoft Excel.

Projects

- **Disaster Tweet Classification Using Deep Learning**
  - Built a text classification model in Python to identify disaster-related tweets using a Kaggle dataset with text, keywords, and location data.
  - Conducted NLP preprocessing (text cleaning, tokenisation) and used pre-trained GloVe embeddings for semantic understanding.
  - Designed a multi-input neural network with bidirectional LSTMs, embedding layers, dropout regularisation, and class weighting to handle imbalanced data.
  - Tuned decision thresholds and evaluated model performance using accuracy (81.9%), F1 score (0.787), and ROC AUC, demonstrating strong model generalisation.
- **Netflix Catalogue Trends Visualisation Using Amazon QuickSight**
  - Analysed and visualised Netflix's catalogue data using Amazon QuickSight to identify trends in content releases and genres.
  - Integrated data from AWS S3 and created interactive dashboards using filters, aggregations, and charts for insight discovery.
  - Gained practical experience in cloud-based business intelligence (BI), data analytics, and data visualisation workflows on AWS.
- **MNIST Digit Classification Using Machine Learning**
  - Developed a digit classification model in Python using the MNIST dataset.
  - Implemented and trained a neural network with TensorFlow and NumPy to recognise handwritten digits.
  - Applied computer vision techniques including image normalisation and reshaping for data preprocessing.
  - Successfully achieved a high model accuracy while gaining hands-on experience with supervised learning and deep learning fundamentals.

Extracurricular Activities and qualifications

- 2nd place, Computer, Electronics and Robotics residential week (University of Southampton, 2022), which was run by the Smallpeice Trust and sponsored by Thales. Teams of 4 had to work together to design, build and code a fully autonomous robot which would compete against other teams in a 'robot wars' style game.
- While at 6th form my friends and I entered the CyberCenturion cyber security competition. Teams had 6 hours to find vulnerabilities and security risks in 3 Linux virtual machines. I did this competition twice and really enjoyed competing while learning a lot about cyber security and Linux operating systems
- I entered Bebras, a problem-solving competition run by the University of Oxford and the Raspberry Pi Foundation and in 2021 came in the top 10% of students in the country in the Elite category which qualified me for the Oxford University Computing Challenge
- Currently studying Machine Learning and Data Science using Kaggle.com and have completed their "introduction to machine learning" and "Intro to Deep Learning" courses.
- Student Member of the British Computer Society
- HSK 1 and HSK 2 Mandarin Exams Passed

Other Education

Bishop Stopford School **September 2021 – June 2023**

Achieved a grade of B in A-Level Maths and C in Physics and Computer Science, along with an Extended Project Qualification (EPQ) titled "Should We Be Afraid of Artificial Intelligence?"